

Fairness-Aware, Adaptive, and Federated Model-Free Scheduling for Hydrogen-Integrated Multi-Area Urban Power and Traffic Networks

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This supplementary document compiles the key modeling assumptions, component efficiencies, cost coefficients, baseline travel-time coefficients, and algorithm parameters used to support the technical analyses reported in the paper.

1) FAPF-TD3 Algorithm

The proposed FAPF-TD3 algorithm uses the following DRL and personalized federated learning settings.

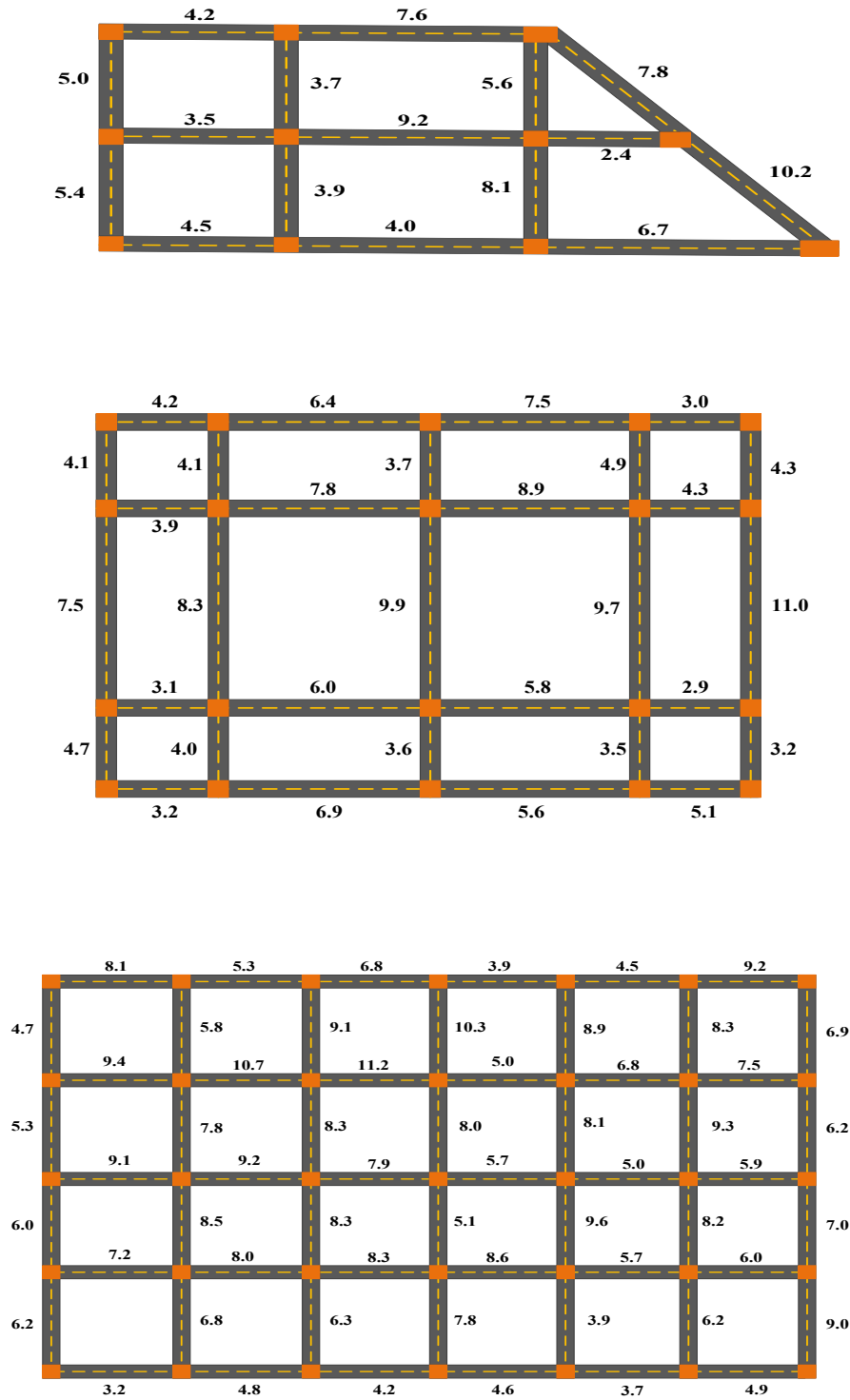
TABLE I
FAPF-TD3 DRL AND FEDERATED LEARNING PARAMETERS

Discount factor	0.99	Global synchronization period	50 episodes
Actor learning rate	3×10^{-4}	Performance smoothing coefficient	0.9
Critic learning rate	3×10^{-4}	Replay buffer size	150,000 transitions
Actor/critic hidden layers	4 layers with 256 neurons each	Area synchronization period	10 episodes
Noise standard deviation	0.10	Maximum interpolation factor	0.8
Target update rate	0.005	Data scale parameter	200
Policy noise clip	0.4	Temperature factor	5.0
Batch size	256	Area preference coefficients	Commercial (2, 30, 1), industrial (2, 10, 5), residential-commercial (5, 30, 1)
Fairness prior	Group equalization	Dynamic fairness range	[0, 1]
Performance weight	0.3	Target policy noise	0.15

2) Non-Congested Travel Time Coefficients

Table II presents the non-congested travel time coefficients for streets and inter-area routes used as baseline inputs in the OD-equilibrium model of the Calgary network. The inter-area coefficients are 62.5 between Areas #1 & #2, 83.7 between Areas #1 & #3, and 106.7 between Areas #2 & #3.

TABLE II
ASSUMED NON-CONGESTED TRAVEL TIME COEFFICIENTS FOR STREETS AND INTER-AREA ROUTES



3) PEM Electrolyzer Parameters

Table III lists the electrochemical and cost coefficients that define the PEM electrolyzer model adopted in this study [1].

TABLE III
ELECTROCHEMICAL AND COST PARAMETERS OF THE PEM ELECTROLYZER

$\mathfrak{I}_1$	$\mathfrak{I}_2$	$\mathfrak{I}_3$	$\mathfrak{I}_4$	$\mathfrak{I}_5$	$\mathfrak{I}_6$	$\mathfrak{I}_7$
99.5%	-9.5788 (m^2/A)	-0.0555 ($\text{m}^2/\text{A} \times ^\circ\text{C}$)	0	1502.7083 (m^4/A)	-70.8005 ($\text{m}^4/\text{A} \times ^\circ\text{C}$)	0
		F	z_{mk}	μ^{PEM}		
		96485.34 (C/mol)	0.002016 (kg/mol)	36 (\$/kg) per year		

4) *Compressor, Hydrogen Tank, and Electrical Storage Parameters*

Table IV summarizes the thermodynamic constants, storage efficiencies, and operating-cost coefficients used to model the hydrogen compressor, hydrogen tank, and on-site electrical storage systems [2]-[4].

TABLE IV
THERMODYNAMIC AND OPERATIONAL PARAMETERS FOR COMPRESSOR,
HYDROGEN TANK, AND ELECTRICAL STORAGE UNITS

Compressor				
ζ	R^{gc}	μ^{comp}	ϱ^{ih}	ϱ^{ie}
1.41	4.124 kJ/(kg·K)	1 (\$/kg) per year	14.31 (kJ/kg.K)	0.8
Hydrogen tank				
μ^{tank}	SoC_{max}^T	SoC_{min}^T	η_{ch}^T	η_{dc}^T
32 (\$/kg) per year	0.9	0.05	0.9	0.9
Electrical storage				
SoC_{min}^{ES}	SoC_{max}^{ES}	η_{ch}^{ES}	η_{dc}^{ES}	μ^{ES}
0.1	0.9	0.95	0.95	0.0005 (\$/kWh)

5) *Hydrogen and EV Charging Pricing Equation Coefficients*

Table V summarizes the key coefficients used in the dynamic hydrogen and EV charging pricing formulations, with values adopted from [5], [6].

TABLE V
COEFFICIENTS USED IN THE HYDROGEN AND EV CHARGING PRICING MODELS

ω_t^{ws}	C_t^{ws}	δ^H	δ^E	η_t^{PEM}
0.6	0.03 (\$/kWh)	1 (\$/kg)	0.1 (\$/kWh)	45 (kWh/kg)

6) *Penalty Coefficients*

TABLE VI
PENALTY COEFFICIENTS

μ^{CH}	μ^{CP}	$\mathfrak{B} \ \& \ \mathfrak{s}$ (Soft constraints penalty & price smoothing coefficients)
130 (\$/kg)	350 (\$/MWh)	100-1000 (\$ per unit) & 0.05

References:

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- [6] Technical Targets for Proton Exchange Membrane Electrolysis [Online]. Accessed: Oct. 20, 2024. Available: [U.S. Department of Energy](https://www.energy.gov/eere/energy-efficiency/technical-targets-for-proton-exchange-membrane-electrolysis).